DHCP Configuration

Q1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

Solution : Configure a DHCP server with two PCs in Cisco Packet Tracer

Steps:

1. In Cisco Packet Tracer add one Server-PT and two PC-PT devices.
2. Connect all three devices to a switch using Copper Straight-Through cables.
3. Click the server and open Services → DHCP.
4. Turn DHCP On and configure the pool as follows:

Setting	**Value**
Pool Name	OfficePool
Default Gateway	192.168.1.1
DNS Server	8.8.8.8
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

6. Configure the server's own static IP as 192.168.1.2 under Desktop → IP Configuration.
7. On each PC, open Desktop → IP Configuration → DHCP.
8. Verify the assigned addresses using Desktop → Command Prompt → ipconfig.

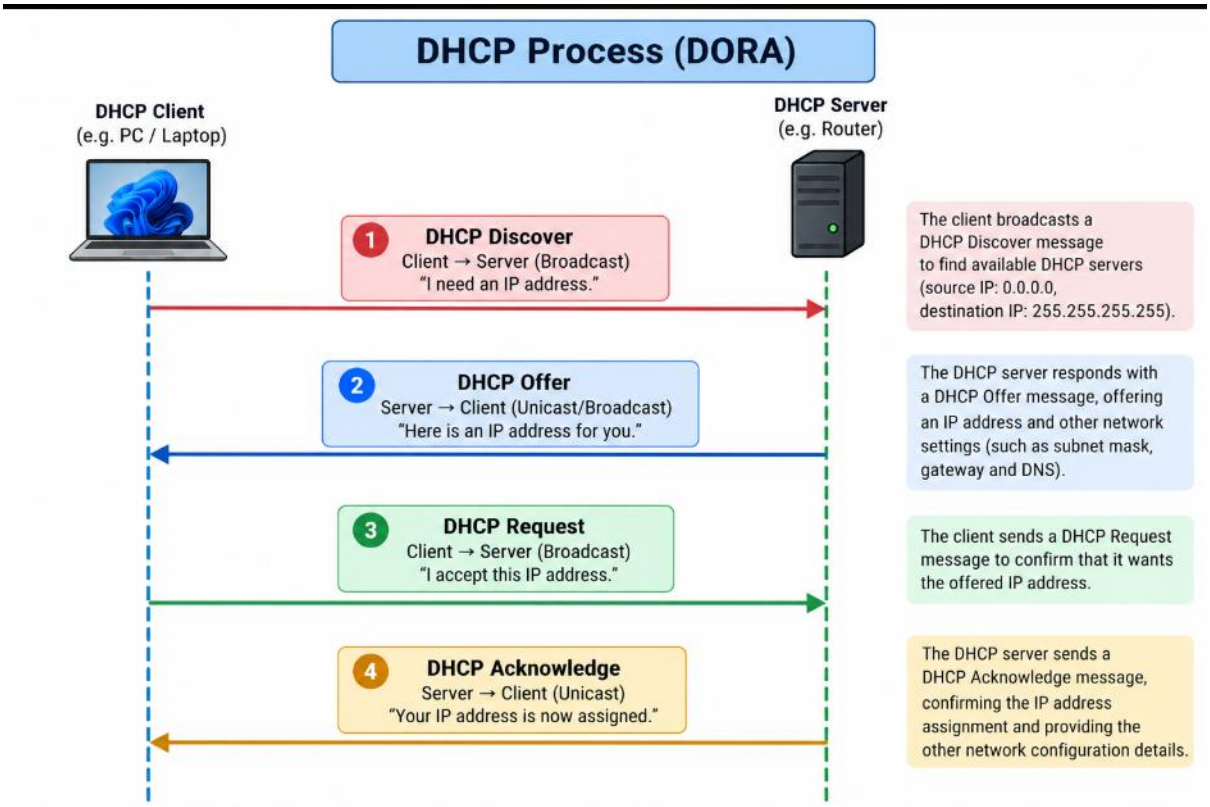
Expected result

Device	**IP Address**
DHCP Server	192.168.1.2 (Static)
PC0	192.168.1.10 (Example)
PC1	192.168.1.11 (Example)

Q2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Hint : Use arrows to show the direction of each message between client and server.

Solution : DHCP process sequence diagram

DHCP uses four steps called DORA: Discover, Offer, Request, Acknowledge.

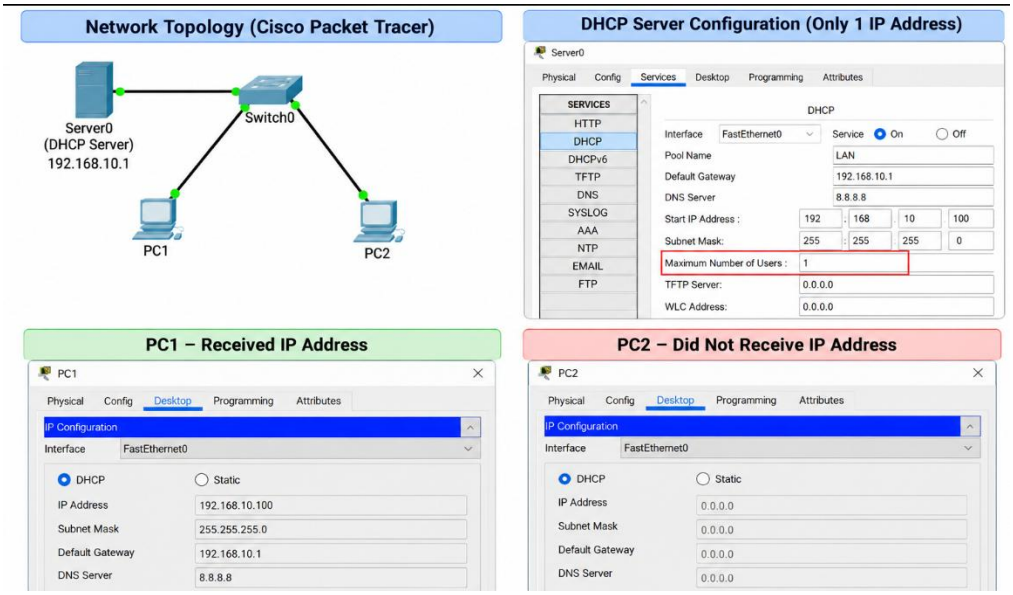


Q3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

Solution : Limit the DHCP pool to one IP address

Steps:

1. Open the DHCP server in Packet Tracer.
2. Go to Services → DHCP.
3. Set the starting IP address to 192.168.1.100.
4. Change Maximum Number of Users to 1.
5. Save the settings.
6. Configure both PCs to obtain their IP addresses using DHCP.
7. Observe the results and take a screenshot.



Expected result

Device	Expected result
PC0	Receives 192.168.1.100
PC1	Cannot obtain an address from the exhausted pool

Explanation: Only one IP address is available in the DHCP pool, so the server can lease it to only one PC at a time. The second PC cannot obtain a new address until an address becomes available.

Q4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

Solution : Simulate a restaurant Wi-Fi network like Zomato

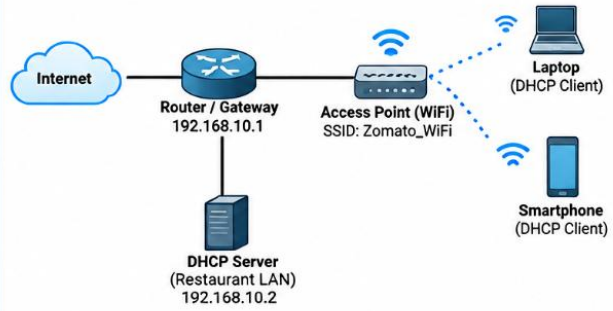
Steps:

1. Add one DHCP server, one switch, one laptop (PC-PT), and one generic PC representing a smartphone.
2. Connect the devices to the switch using Copper Straight-Through cables.
3. Assign the server a static IP address: 192.168.10.2.
4. Open Server → Services → DHCP.
5. Enable DHCP and configure the pool:

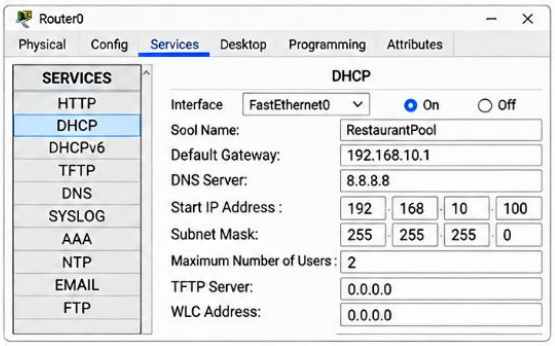
Setting	**Value**
Pool Name	RestaurantWiFi
Default Gateway	192.168.10.1
DNS Server	8.8.8.8
Start IP Address	192.168.10.100
Subnet Mask	255.255.255.0
Maximum Users	50

6. Open each client and select Desktop → IP Configuration → DHCP.
7. Check the assigned IP addresses.

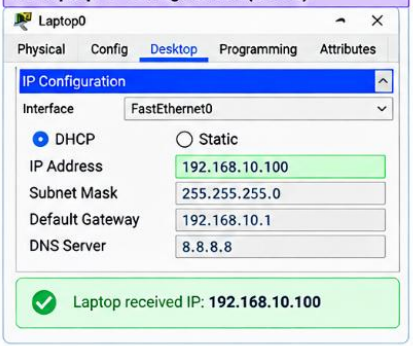
1. Network Topology (Zomato-style Restaurant WiFi)



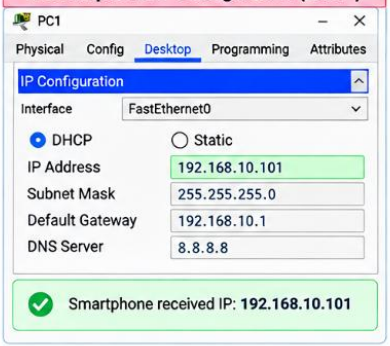
2. DHCP Server Configuration (Router / Server)



3. Laptop IP Configuration (DHCP)



4. Smartphone IP Configuration (DHCP)



5. Why DHCP is Useful?

- Automatically assigns IP addresses to devices (laptop, smartphone) without manual configuration.
- Prevents IP conflicts by managing the available address pool.
- Makes it easy for customers to connect to the restaurant WiFi (like Zomato-style) without technical knowledge.

Result:

Both the laptop and the smartphone successfully received IP addresses from the DHCP server, allowing them to connect to the restaurant WiFi (Zomato_WiFi) and access the internet.